

Research Paper Draft

Title: Geospatial Assessment of Land Surface Temperature, Vegetation, Water Bodies and Land Use/Land Cover in Purulia District, West Bengal Using Google Earth Engine and Landsat 8

Abstract

Land Surface Temperature (LST), vegetation, water resources and land use/land cover (LULC) were assessed for Purulia District using Landsat 8 and Google Earth Engine. LST ranged from 19.58°C to 40.16°C (mean 28.17°C). NDVI ranged from -0.195 to 0.451 (mean 0.199), while NDWI ranged from -0.414 to 0.201 (mean -0.205). LULC analysis showed cropland (52.76%) as the dominant class, followed by tree cover (27.25%) and grassland (16.45%). The workflow demonstrates the usefulness of Google Earth Engine for district-level environmental assessment.

Objectives

- Prepare a Land Surface Temperature map.
- Assess vegetation using NDVI.
- Assess surface water using NDWI.
- Map Land Use/Land Cover using ESA WorldCover.
- Support environmental planning using geospatial analysis.

Key Results

Parameter	Minimum	Maximum	Mean
LST (°C)	19.58	40.16	28.17
NDVI	-0.195	0.451	0.199
NDWI	-0.414	0.201	-0.205

LULC Statistics

Land Cover	Area (km ²)	Percentage (%)
Cropland	3301.04	52.76
Tree Cover	1705.09	27.25
Grassland	1029.23	16.45
Built-up	93.61	1.50
Water Bodies	88.10	1.41
Bare/Sparse Vegetation	27.00	0.43
Shrubland	12.54	0.20
Herbaceous Wetland	0.11	0.00

Conclusion

The integrated analysis of LST, NDVI, NDWI and LULC provides a comprehensive understanding of environmental conditions in Purulia District. The workflow can be extended to multi-year change detection and correlation analysis.

1. Introduction

Land Surface Temperature (LST) is one of the most important environmental indicators used to evaluate climate variability, land degradation, urban expansion, and ecosystem health. Changes in land use and land cover (LULC) directly influence surface temperature by altering vegetation density, soil moisture, and surface characteristics. Remote sensing and Geographic Information System (GIS) techniques provide an efficient approach for monitoring these environmental changes over large geographical areas.

Google Earth Engine (GEE) has emerged as a powerful cloud-based geospatial platform for processing large satellite datasets. It provides free access to Landsat, Sentinel, MODIS, and other datasets, enabling rapid analysis without the need for high-performance computers. Landsat 8 Collection 2 Level-2 products provide multispectral and thermal information suitable for calculating Land Surface Temperature (LST), Normalized Difference Vegetation Index (NDVI), and Normalized Difference Water Index (NDWI).

Purulia District, located in the western part of West Bengal, is characterized by a semi-arid climate, plateau topography, seasonal rivers, and dry deciduous forests. Agriculture is the major occupation, while increasing land-use changes and climatic variability have affected environmental conditions in the district. Therefore, an integrated geospatial assessment is important for understanding the spatial relationship among temperature, vegetation, water resources, and land use.

The present study integrates Google Earth Engine and QGIS to generate LST, NDVI, NDWI, and LULC maps for Purulia District and evaluates their statistical characteristics for environmental assessment.

2. Study Area

Purulia District is situated in the western part of West Bengal and forms a part of the Chota Nagpur Plateau. The district extends over approximately **6,259 km²** and experiences a tropical climate with hot summers and moderate monsoon rainfall. The landscape is characterized by undulating terrain, lateritic soils, dry deciduous forests, and seasonal rivers.

Agriculture is the dominant land use, while forest cover is concentrated in the Ajodhya Hills and surrounding plateau regions. Because of its climatic variability and recurring water scarcity, Purulia is considered an important region for environmental and climate-related studies.

3. Data and Methodology

The study utilized Landsat 8 Collection 2 Level-2 satellite imagery processed in Google Earth Engine. Administrative boundary data were obtained from the GADM database, while the ESA WorldCover 2021 dataset was used for land use/land cover mapping.

The methodology consisted of the following steps:

- Collection of Landsat 8 imagery
- Clipping of satellite data using the Purulia district boundary
- Generation of LST using the thermal band (ST_B10)
- Calculation of NDVI and NDWI using spectral indices
- Extraction of LULC from ESA WorldCover
- Statistical analysis using Google Earth Engine
- Preparation of final maps in QGIS

The processed raster layers were exported in GeoTIFF format and visualized in QGIS. Statistical summaries were extracted using the `reduceRegion()` function.

4. Results and Discussion

The generated LST map showed considerable spatial variation across Purulia District. Surface temperatures ranged from **19.58°C to 40.16°C**, with an average temperature of **28.17°C**. Lower temperatures were observed in forested regions, whereas relatively higher temperatures occurred over croplands, built-up areas, and sparsely vegetated land.

The NDVI values ranged from **-0.195 to 0.451**, with an average value of **0.199**, indicating moderate vegetation conditions. Higher NDVI values were mainly associated with the forested western and south-western parts of the district.

NDWI values ranged between **-0.414 and 0.201**, with an average value of **-0.205**. The predominance of negative NDWI values indicates comparatively limited surface water availability during the selected period.

LULC analysis revealed that cropland occupied **52.76%** of the district area, followed by tree cover (**27.25%**) and grassland (**16.45%**). Built-up land accounted for only **1.50%**, indicating that Purulia remains predominantly rural in nature.

5. Correlation Analysis

Pearson correlation analysis was carried out to evaluate the relationship between vegetation and land surface temperature.

The analysis produced a **weak negative correlation ($r = -0.210$)** with a **highly significant p-value ($p < 0.001$)**.

This indicates that areas with relatively higher vegetation cover generally exhibit lower land surface temperatures. However, the relatively weak correlation suggests that additional factors such as land use, elevation, soil moisture, and topographic characteristics also influence the thermal environment of Purulia District.

6. Conclusion

The present study successfully integrated Google Earth Engine and QGIS for environmental assessment of Purulia District.

The generated LST, NDVI, NDWI, and LULC maps clearly demonstrate the spatial variability of environmental conditions across the district. Cropland was identified as the dominant land cover, while forested areas exhibited comparatively lower land surface temperatures. The statistically significant negative relationship between NDVI and LST confirms the cooling effect of vegetation on the land surface.

The adopted methodology is simple, efficient, and suitable for district-level environmental monitoring. The workflow can easily be extended for multi-temporal change detection and climate change assessment.

Future Scope

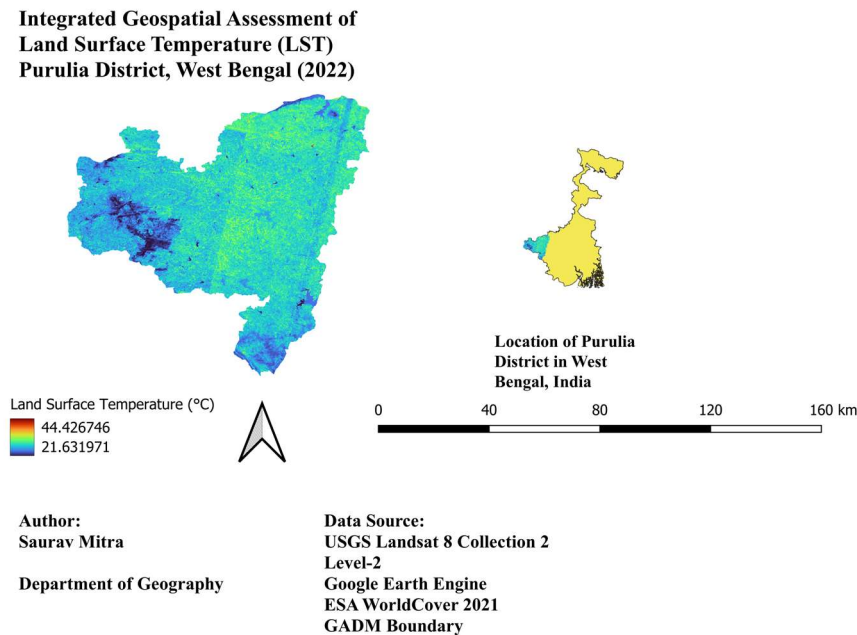
Future studies may include:

- Multi-year LST analysis (2015–2025)
- Urban Heat Island assessment
- Machine Learning-based LULC classification
- Climate change hotspot mapping
- Integration of Sentinel-2 imagery
- Accuracy assessment of LULC classification

References (APA Style)

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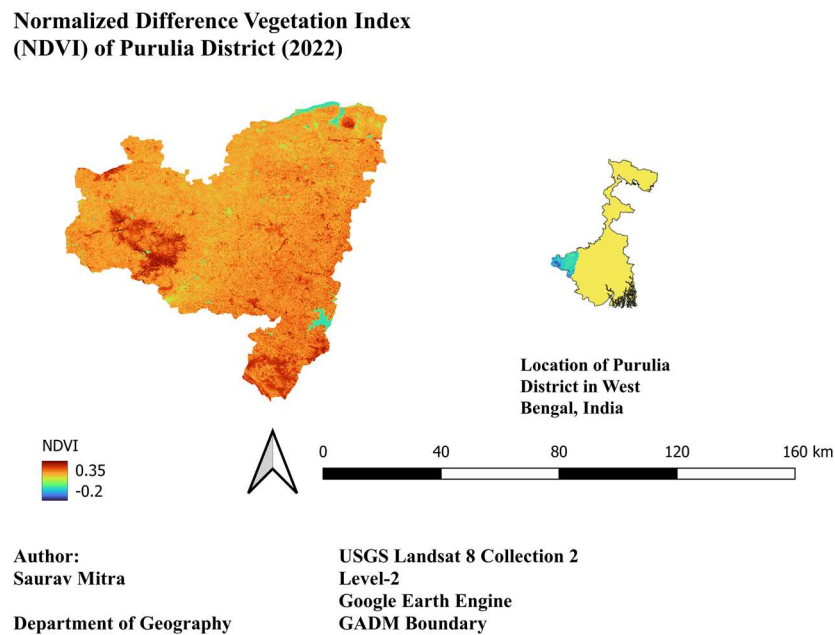
Figure 1: Land Surface Temperature (LST) of Purulia District, West Bengal (2022).



Explanation:

The LST map indicates considerable spatial variation in surface temperature across Purulia District. Surface temperature ranged from **19.58°C to 40.16°C**, with a mean value of **28.17°C**. Lower temperatures are mainly associated with forested regions, whereas higher temperatures occur over cropland, barren land, and built-up areas.

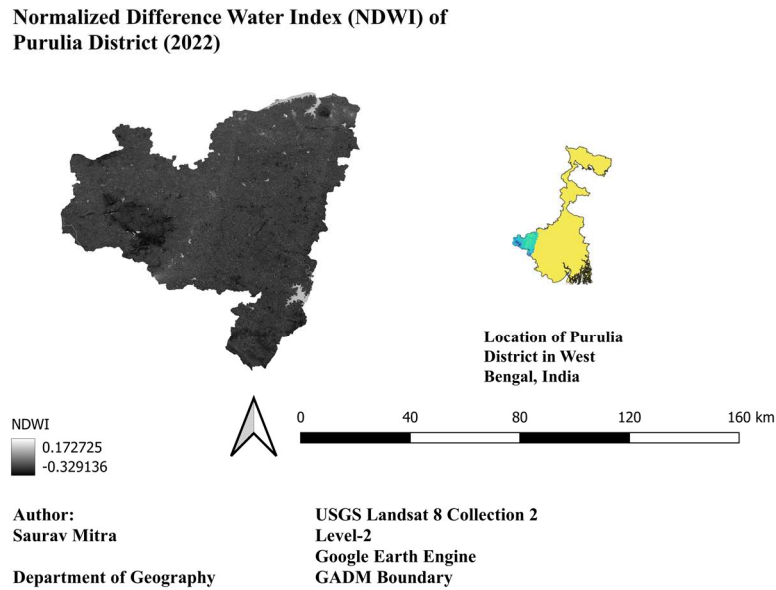
Figure 2: Normalized Difference Vegetation Index (NDVI) of Purulia District (2022).



Explanation:

The NDVI map represents the spatial distribution of vegetation cover. NDVI values ranged from **-0.195** to **0.451**, with an average value of **0.199**. Higher NDVI values indicate dense vegetation, while lower values represent sparse vegetation, exposed soil, or water bodies.

Figure 3: Normalized Difference Water Index (NDWI) of Purulia District (2022).

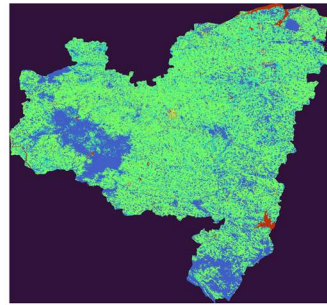


Explanation:

The NDWI map highlights the distribution of surface water within the district. NDWI values ranged between **-0.414** and **0.201**, with a mean value of **-0.205**. Positive values indicate water bodies, while negative values represent non-water surfaces such as vegetation, cropland, and barren land.

Figure 4: Land Use/Land Cover (LULC) of Purulia District (2021).

**Integrated Geospatial Assessment of
Land Surface Temperature (LST)
Purulia District, West Bengal (2022)**



**Location of Purulia
District in West
Bengal, India**

Land Use / Land Cover
90
0



0 40 80 120 160 km

Author:
Saurav Mitra

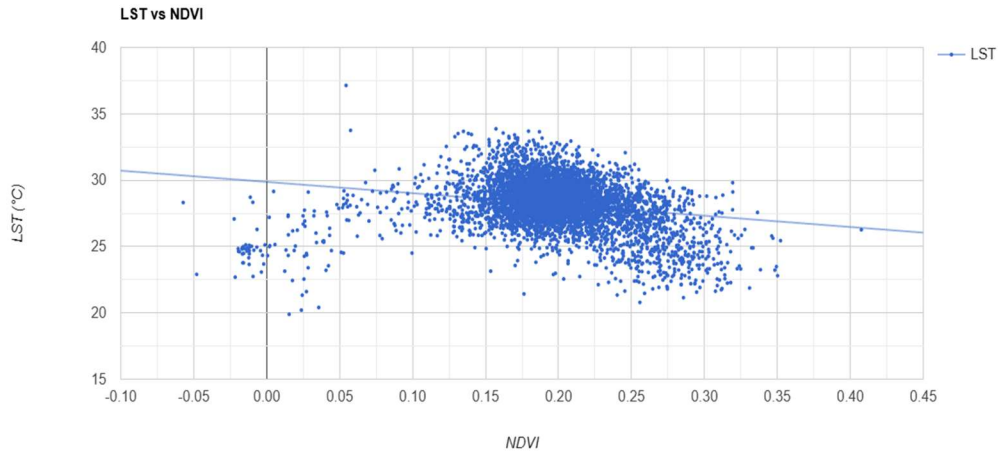
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**ESA WorldCover 2021
Google Earth Engine
GADM Boundary**

Explanation:

The LULC map shows that cropland is the dominant land cover, occupying **52.76%** of the district, followed by tree cover (**27.25%**) and grassland (**16.45%**). Built-up areas account for only **1.50%**, indicating that Purulia is predominantly rural with extensive agricultural and forest landscapes.

Figure 5: Relationship between Land Surface Temperature (LST) and NDVI.



Explanation:

The scatter plot illustrates a **weak but statistically significant negative correlation** between LST and NDVI ($r = -0.210$, $p < 0.001$). This indicates that areas with relatively higher vegetation cover tend to have lower land surface temperatures, although additional factors such as land use, topography, and soil moisture also influence the thermal environment.